

SIMATS ENGINEERING
ASSESSMENT TOOL 2
Case study

COURSE NAME: Computer Networks

COURSE CODE: CSA0720

NAME : VARSHA VARDHINI.S

COURSE OUTCOME COVERED

CO3: Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions

Assessment Tool Weightage: 40%

Case study

88/100

QUESTIONS:4

Total Marks:40

1.A multinational IT company uses a cloud-based video conferencing platform for daily client meetings. During peak business hours, employees experience frequent voice distortion, frozen video frames, and delayed communication. Network monitoring reports indicate an **8% packet loss**, **180 ms jitter**, and fluctuating bandwidth utilization. The IT administrator suspects that the issue is related to the transport protocol or application-level buffering strategy.

Questions

1. Analyze whether the problem is primarily caused by the **transport protocol (TCP/UDP)** or the **application layer**.
2. Justify your answer based on the communication requirements of real-time multimedia applications.
3. Recommend suitable protocol-level or application-level improvements to enhance conferencing quality.

2.A multinational software company hosts its web services across multiple continents. Employees frequently report long delays before the application homepage loads. Network analysis shows that the **average DNS lookup time is approximately 800 ms**, significantly affecting user experience. The

organization plans to redesign its DNS infrastructure while ensuring continuous service availability during server failures.

Questions

1. Analyze the reasons behind the high DNS resolution time.
2. Propose a suitable DNS architecture using techniques such as **Anycast DNS**, **DNS pre-fetching**, and **TTL optimization** to reduce the lookup time below **50 ms**.
3. Evaluate the advantages and limitations of your proposed architecture in terms of performance, scalability, and availability.

3.A cloud service provider delivers applications to millions of users worldwide. During promotional events, users experience intermittent delays in accessing cloud-hosted services. Investigation reveals that DNS queries are taking nearly **800 ms** because all requests are directed to a single primary DNS server. The company wants a highly available DNS solution capable of handling global traffic efficiently.

Questions

4. Examine the impact of centralized DNS architecture on application performance.
5. Design a distributed DNS solution that minimizes lookup latency while maintaining high availability.
6. Analyze how **TTL values**, **Anycast routing**, and **DNS caching** influence system performance and fault tolerance.

4.A financial institution operates an electronic stock trading platform where thousands of buy and sell orders are submitted every second. During market opening hours, the **average order acknowledgment latency increases from 1 ms to 40 ms**, resulting in delayed trade confirmations and customer dissatisfaction. Network engineers observe increased TCP retransmissions, longer connection establishment times, and excessive buffering.

Questions

1. Analyze three possible TCP-related factors responsible for the latency increase, considering **flow control**, **Nagle's algorithm**, and **SYN queue limitations**.
2. Explain how each factor affects transaction processing in high-frequency systems

3. Recommend appropriate TCP configuration changes to reduce latency and improve transaction reliability.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues <i>2.5</i>	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply <i>2.5</i>	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
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Clarity	10	Clear, well-structured, and easy to understand <i>1</i>	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

MARK ALLOCATION

	Understanding of Case (25%)	Application of Concepts (25%)	Analysis (20%)	Solution Design (20%)	Clarity (10%)	
Q1	<i>2</i>	<i>2</i>	<i>2</i>	<i>1.5</i>	<i>1</i>	<i>8.5</i>
Q2	<i>2</i>	<i>2</i>	<i>1.5</i>	<i>2</i>	<i>0.5</i>	<i>8</i>
Q3	<i>2.5</i>	<i>2</i>	<i>1.5</i>	<i>2</i>	<i>0.5</i>	<i>8.5</i>
Q4	<i>2.5</i>	<i>2.5</i>	<i>2</i>	<i>2</i>	<i>1</i>	<i>10</i>
Q5						

35
40

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CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

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1.A multinational IT company uses a cloud-based video conferencing platform for daily client meetings. During peak business hours, employees experience frequent voice distortion, frozen video frames, and delayed communication. Network monitoring reports indicate an **8% packet loss, 180 ms jitter**, and fluctuating bandwidth utilization. The IT administrator suspects that the issue is related to the transport protocol or application-level buffering strategy.

Questions

1. Analyze whether the problem is primarily caused by the **transport protocol (TCP/UDP)** or the **application layer**.
2. Justify your answer based on the communication requirements of real-time multimedia applications.
3. Recommend suitable protocol-level or application-level improvements to enhance conferencing quality.

ANSWER :

Analyze whether the problem is primarily caused by the transport protocol (TCP/UDP) or the application layer.

The problem is primarily caused by network conditions affecting the transport layer, rather than the application layer alone. The 8% packet loss can cause missing audio/video packets, while 180 ms jitter causes irregular packet arrival, leading to voice distortion and frozen frames. Fluctuating bandwidth further worsens video quality. The application layer may also contribute if its jitter buffer cannot adapt properly to these changing conditions.

Justify based on the communication requirements of real-time multimedia applications.

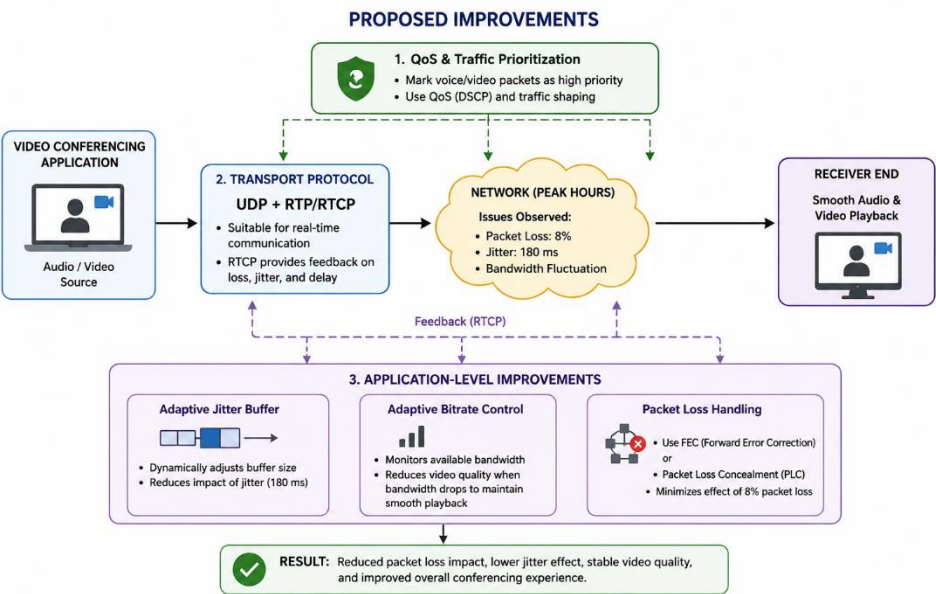
Real-time multimedia applications require low latency, low jitter, and continuous data delivery. Similar to real-world conferencing platforms such as [Zoom](#), UDP is generally preferred because TCP retransmits lost packets, which can increase delay and cause further freezing. Therefore, UDP is suitable for conferencing, but high packet loss and jitter can still significantly affect quality. Hence, the issue is mainly due to poor network performance, not the choice of UDP itself.

Recommend suitable protocol-level or application-level improvements.

The company can improve conferencing quality by:

- Applying QoS to prioritize voice and video traffic during peak hours.
- Using UDP with RTP/RTCP for real-time media transmission and network feedback.
- Implementing an adaptive jitter buffer to handle the high jitter.
- Using adaptive bitrate control to reduce video quality when bandwidth decreases.

3. Recommend suitable protocol-level or application-level improvements.



Conclusion: The case is mainly a network/transport performance issue, supported by possible application-layer buffering limitations. Using QoS and adaptive real-time communication techniques would provide a feasible solution.

2.A multinational software company hosts its web services across multiple continents. Employees frequently report long delays before the application homepage loads. Network analysis shows that the **average DNS lookup time is approximately 800 ms**, significantly affecting user experience. The organization plans to redesign its DNS infrastructure while ensuring continuous service availability during server failures.

Questions

- 1. Analyze the reasons behind the high DNS resolution time.
- 2. Propose a suitable DNS architecture using techniques such as **Anycast DNS**, **DNS pre-fetching**, and **TTL optimization** to reduce the lookup time below **50 ms**.

ANSWER :

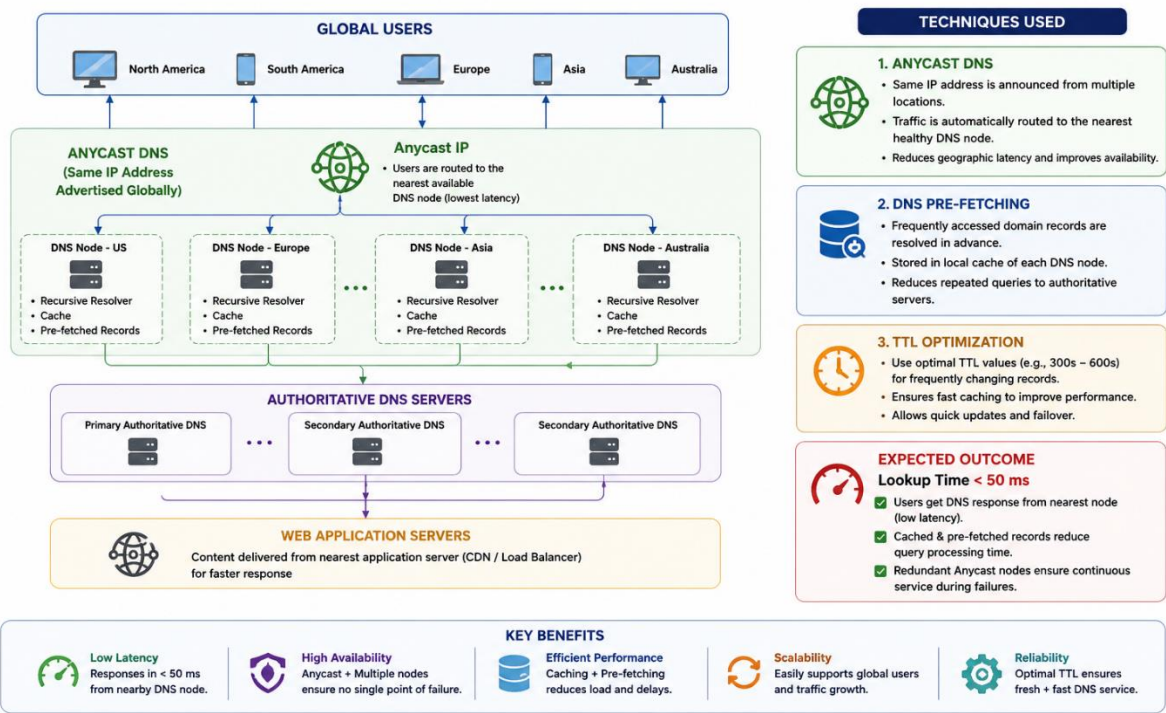
Analyze the reasons behind the high DNS resolution time

The 800 ms DNS lookup time is unusually high and can be caused by:

- Centralized or geographically distant DNS servers: Users across multiple continents may need to contact a DNS server located far away, increasing latency.
- Multiple recursive DNS queries: If records are not cached, the resolver may need to contact root, TLD, and authoritative DNS servers.
- High TTL misconfiguration: Very high or poorly managed TTL values can delay updates and failover, while extremely low TTL values can increase repeated DNS queries.
- Server overload or failure: A single DNS server can become a bottleneck during high traffic, increasing response time and reducing availability.
- Lack of caching/pre-fetching: Frequently accessed domain records may be repeatedly resolved instead of being available from cache.

Thus, the main issue is likely a **centralized and insufficiently optimized DNS architecture**, causing both high lookup latency and poor fault tolerance.

2. Propose a suitable DNS architecture using techniques such as Anycast DNS, DNS pre-fetching, and TTL optimization to reduce the lookup time below 50 ms.



Recommended Techniques

- Anycast DNS: Multiple DNS servers use the same IP address, and users are automatically directed to the nearest available DNS node, reducing geographical delay.
- DNS Pre-fetching: Frequently requested domain records are resolved in advance and stored in cache, reducing repeated lookup time.
- TTL Optimization: Use an appropriate TTL to balance fast DNS caching with the ability to update records and perform failover quickly.
- Redundant DNS Nodes: Deploy DNS servers across multiple geographical regions. If one server fails, requests are automatically handled by another node.

3. Evaluate the advantages and limitations of your proposed architecture in terms of performance, scalability, and availability.

3.A cloud service provider delivers applications to millions of users worldwide. During promotional events, users experience intermittent delays in accessing cloud-hosted services. Investigation reveals that DNS queries are taking nearly **800 ms** because all requests are directed to a single primary DNS server. The company wants a highly available DNS solution capable of handling global traffic efficiently.

Questions

4. Examine the impact of centralized DNS architecture on application performance.
5. Design a distributed DNS solution that minimizes lookup latency while maintaining high availability.
6. Analyze how **TTL values**, **Anycast routing**, and **DNS caching** influence system performance and fault tolerance.

ANSWER :

Examine the impact of centralized DNS architecture on application performance

In this case, all DNS queries are directed to a single primary DNS server, creating a single point of failure and performance bottleneck. During promotional events, millions of users generate DNS requests simultaneously, causing server overload and increasing lookup time to nearly 800 ms.

A centralized DNS architecture results in:

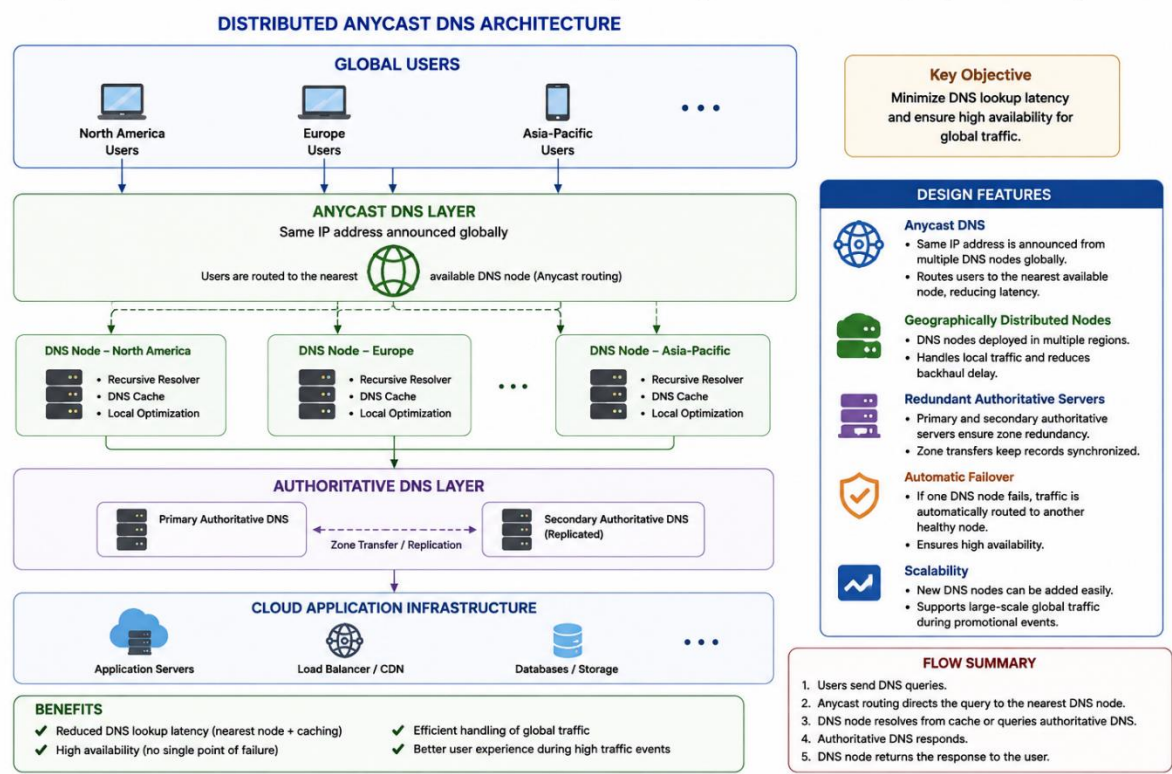
- High latency for geographically distant users.
- Server congestion during heavy traffic.
- A single point of failure if the primary DNS server becomes unavailable.
- Delayed access to cloud applications, reducing user experience.

Thus, centralized DNS is not suitable for a globally distributed application with high traffic.

Design a distributed DNS solution

The company should use a **distributed Anycast DNS architecture** with DNS nodes deployed across multiple geographical regions

2. Design a distributed DNS solution that minimizes lookup latency while maintaining high availability.



Design features:

- Anycast routes users to the **nearest available DNS server**.
- Multiple DNS nodes provide **load distribution**.
- Primary and secondary authoritative servers provide **redundancy**.
- If one DNS node fails, traffic is automatically routed to another available node.

This architecture reduces lookup latency and improves availability.

Analyze the influence of TTL, Anycast routing, and DNS caching :

Technique	Effect on Performance	Effect on Fault Tolerance
TTL Optimization	Appropriate TTL reduces repeated DNS lookups and improves response time.	Lower TTL allows faster DNS updates and failover.
Anycast Routing	Routes users to the nearest DNS node, reducing geographical latency.	If one node fails, traffic can be routed to another node.
DNS Caching	Cached records reduce repeated queries to	Cached records can temporarily maintain access, but stale records

	authoritative servers, lowering load and delay.	may delay changes if TTL is too high.
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Conclusion

A distributed DNS architecture using Anycast, geographically distributed DNS nodes, optimized TTL values, and DNS caching is more suitable than a centralized architecture. It reduces the 800 ms lookup delay, distributes global traffic, and ensures high availability even when individual DNS servers fail.

4.A financial institution operates an electronic stock trading platform where thousands of buy and sell orders are submitted every second. During market opening hours, the **average order acknowledgment latency increases from 1 ms to 40 ms**, resulting in delayed trade confirmations and customer dissatisfaction. Network engineers observe increased TCP retransmissions, longer connection establishment times, and excessive buffering.

Questions

- 1. Analyze three possible TCP-related factors responsible for the latency increase, considering **flow control**, **Nagle's algorithm**, and **SYN queue limitations**.
- 2. Explain how each factor affects transaction processing in high-frequency systems
- 3. Recommend appropriate TCP configuration changes to reduce latency and improve transaction reliability.

ANSWER :

Analyze three TCP-related factors responsible for the latency increase

a) Flow Control:

TCP flow control uses the receiver window to prevent a fast sender from overwhelming the receiver. During market opening, thousands of orders arrive simultaneously. If the receiver application processes data slowly, the receive window may shrink, forcing the sender to wait and increasing order acknowledgment latency.

b) Nagle's Algorithm:

Nagle's algorithm combines small packets before transmission to reduce network overhead. However, stock trading transactions are usually small and time-sensitive. Waiting for packets to be combined can introduce additional delay, contributing to the increase from **1 ms to 40 ms**.

c) **SYN Queue Limitations :**

During heavy traffic, a large number of clients may attempt TCP connections simultaneously. If the server's **SYN queue becomes full**, new connection requests may be delayed or dropped, resulting in longer connection establishment times and possible retransmissions.

Effect on Transaction Processing in High-Frequency Systems

TCP Factor	Effect on Transaction Processing
Flow Control	A reduced receive window slows the sender, causing orders and acknowledgments to wait.
Nagle's Algorithm	Small order packets may be delayed while waiting to be combined, increasing transaction latency.
SYN Queue Limitation	Connection requests may be delayed or dropped, increasing connection setup time and TCP retransmissions.

Recommended TCP Configuration Changes

- Optimize TCP receive and send buffer sizes to prevent excessive buffering while maintaining sufficient throughput.
- Enable TCP_NODELAY to disable Nagle's algorithm for small, latency-sensitive order messages.
- Increase the SYN backlog/SYN queue size to handle the large number of simultaneous connection requests.
- Use persistent or long-lived TCP connections to avoid repeated connection establishment during market opening.
- Tune TCP retransmission and timeout parameters carefully to reduce unnecessary delays while maintaining reliable delivery.
- Apply traffic prioritization and load balancing to distribute incoming trading traffic across multiple servers.

Conclusion :

The latency increase is mainly caused by a combination of **flow-control delays, Nagle's algorithm delaying small packets, and SYN queue congestion during heavy connection requests**. By tuning TCP buffers, enabling TCP_NODELAY, increasing the SYN backlog, and reusing persistent connections, the trading platform can reduce acknowledgment latency while maintaining TCP reliability.

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